GROMM goes Production

Generative Risk-neutral Option Market Models II - Production Grade

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Abstract

We refine the GROMM model presented in [WBH26] such that it has an expanding strike and expory grid, and its parameters can be estimated based on prevailing grid geometry. We compare LSTM, continuous-time GRU, and compact Mamba-style selective state-space backbones for its discrete local-volatility factor law, including variants with state-dependent spot–volatility correlation.

The resulting model has a near-perfect fit to observed option prices and outperforms comparable models in $\mathbb{P}$-measure simulations.

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1 Introduction

2 Factor-Law Evaluation and Conditional Arbitrage Check

2.1 Out-of-sample Factor-Law Evaluation

The historical sample is too short to justify selecting a conditional factor law from its in-sample fit. We therefore use a rolling-origin experiment. The experiment contains four consecutive test blocks of 63 business days, for a total of 252 out-of-fold observations from 13 August 2024 to 28 August 2025. Before each block, a model is fitted using only earlier observations. The number

of recurrent training epochs is selected on the final 20% of that earlier sample, after which the model is refitted on the entire pre-block history.

Let $Z_t \in \mathbb{R}^F$ denote the observed factor innovation on day t, and let $C_{m,t}$ be the conditional covariance forecast by model m. The primary $\mathbb{Q}$-measure diagnostic is the Gaussian negative log score per factor and day,

$$L^{\mathbb{Q}}_{m,t} := \frac{1}{2F}\left(\log\det C_{m,t} + Z_t^\top C_{m,t}^{-1} Z_t\right). \tag{1}$$

This is a proper scoring rule for a zero-mean conditional Gaussian factor law. The original models use diagonal $C_{m,t} = \mathrm{diag}(s^2_{m,t,1}, \ldots, s^2_{m,t,F})$. To test state-dependent spot–volatility correlation without estimating an unrestricted loading matrix, we also fit

$$C_{m,t} = D_{m,t}\begin{pmatrix} 1 & \rho_{m,t}^\top \\ \rho_{m,t} & I \end{pmatrix} D_{m,t}, \qquad \rho_{m,t} = \bar\rho \frac{u_{m,t}}{\sqrt{1+\|u_{m,t}\|_2^2}}, \qquad \bar\rho = 0.95, \tag{2}$$

where the first factor is the Gaussianized spot shock and the remaining factors are its unconditionally orthogonal DLV residuals. Hence $\|\rho_{m,t}\|_2 < \bar\rho$ and the covariance is positive definite by construction. Through the fixed DLV loading basis, changing $\rho_{m,t}$ changes both node-level spot–DLV correlations and the effective conditional loading directions.

Performance is reported relative to the AR(1) benchmark as the paired daily improvement

$$\Delta^{\mathbb{Q}}_m := \frac{1}{T}\sum_{t=1}^{T}\left(L^{\mathbb{Q}}_{\mathrm{AR}(1),t} - L^{\mathbb{Q}}_{m,t}\right). \tag{3}$$

Thus, positive values indicate an improvement over AR(1). Confidence intervals resample complete days using a moving-block bootstrap with ten-business-day blocks and 20,000 replications. The conclusions below are unchanged with block lengths of five or twenty business days.

For the physical drift, let $b^{\mathbb{P}}_{t,f}$ denote the estimated coefficient of the attainable $\mathbb{P}$-drift in factor direction f, let $\widehat{b}^{\mathbb{P}}_{m,t,f}$ be its forecast, and let τ_f be its training sample standard deviation. We report

$$L^{\mathbb{P}}_{m,t} := \frac{1}{F}\sum_{f=1}^{F}\left(\frac{b^{\mathbb{P}}_{t,f} - \widehat{b}^{\mathbb{P}}_{m,t,f}}{\tau_f}\right)^2. \tag{4}$$

Unlike (1), this evaluates an estimated attainable drift target. It is not a realized hedging profit-and-loss statistic.

2.1.1 LSTM, continuous-time GRU, and Mamba-style factor laws

All three recurrent models receive the same causal input x_t: the standardized current DLV-level coordinates concatenated with the previous day's factor innovation. The latter is set to zero whenever the historical panel is not contiguous. From a common hidden state, the output layer predicts the F attainable $\mathbb{P}$-drift coefficients and the F conditional log-scales. The correlated variants additionally predict the $F-1$ unconstrained coordinates u_t in (2). Thus the models differ only in their state update, not in their information set or output parameterization.

For the LSTM, with σ denoting the logistic function, we use

$$(i_t, f_t, g_t, o_t) = \big(\sigma, \sigma, \tanh, \sigma\big)\left(W_x x_t + W_h h_{t-1} + b\right), \tag{5}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t, \tag{6}$$

$$h_t = o_t \odot \tanh(c_t), \qquad y_t = W_o h_t + b_o. \tag{7}$$

The hidden and cell states are reset at a break in the daily sequence.

The continuous-time GRU uses a conventional reset-gated candidate but replaces the usual sigmoid update gate by an exponential decay over the actual time step. Specifically,

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r), \qquad \lambda_t = \mathrm{softplus}(W_\lambda x_t + U_\lambda h_{t-1} + b_\lambda), \tag{8}$$

$$\widetilde{h}_t = \tanh(W_c x_t + U_c(r_t \odot h_{t-1}) + b_c)\,, \qquad a_t = \exp(-\lambda_t \Delta t), \tag{9}$$

$$h_t = a_t \odot h_{t-1} + (1 - a_t) \odot \widetilde{h}_t, \qquad y_t = W_o h_t + b_o. \tag{10}$$

Here $\Delta t = 1/252$ on the daily panel. The annual decay is initialized at 25, corresponding to one-day retention $\exp(-25/252) = 0.906$, and is then learned as a positive state-dependent quantity.

The model denoted "Mamba" in the experiments is a compact Mamba-style selective state-space recurrence rather than a full stacked Mamba3 network. Its input-dependent retention gate and candidate state are

$$a_t = \sigma(W_a x_t + b_a), \qquad \widetilde{h}_t = \tanh(W_c x_t + b_c), \tag{11}$$

$$h_t = a_t \odot h_{t-1} + (1 - a_t) \odot \widetilde{h}_t, \qquad y_t = W_o h_t + b_o. \tag{12}$$

This has fewer recurrent interactions than the LSTM and provides a useful small-sample test of whether input-dependent memory is sufficient. All three models use 32 hidden coordinates. Their output vector is partitioned as

$$y_t = \left(\widehat{b}_t^{\mathbb{P}}, \ell_t, u_t\right), \qquad s_{t,f} = \bar{s}_f \exp(\mathrm{clip}(\ell_{t,f}, -\log 3, \log 3))\,, \tag{13}$$

where the final block is omitted in the diagonal models. On each fitting prefix we minimize

$$\frac{1}{T_{\text{train}}} \sum_t L^{\mathbb{Q}}_{m,t} + \frac{1}{T_{\text{train}}} \sum_t L^{\mathbb{P}}_{m,t} + \lambda_\rho \frac{1}{T_{\text{train}}} \sum_t \|\rho_{m,t}\|_2^2, \qquad \lambda_\rho = 10^{-2}, \tag{14}$$

using Adam. The correlation penalty is zero for the diagonal variants. Training duration is selected using only the pre-test validation suffix and the selected number of epochs is then used to refit the complete pre-test history.

Table 1: Rolling-origin factor-law evaluation. Lower mean scores are better. Improvements and confidence intervals are measured relative to AR(1); positive improvements are favourable. The reported $\|\rho_t\|_2$ is a latent loading norm, not an observable spot–volatility correlation.

	$\mathbb{Q}$ factor covariance				Attainable $\mathbb{P}$ drift		
Model	Mean score	Improvement	95% interval	Mean $\|\rho_t\|_2$	Mean error	Improvement	95% interval
Fixed	0.5258	−0.0106	[−0.0258, 0.0003]	0	0.7342	−0.3270	[−0.4153, −0.2478]
AR(1)	0.5152	—	—	0	0.4072	—	—
LSTM	0.5063	0.0088	[−0.0471, 0.0658]	0	**0.2722**	**0.1350**	[**0.0993**, **0.1682**]
LSTM with ρ_t	0.6125	−0.0973	[−0.3397, 0.0475]	0.9007	0.2927	0.1145	[0.0663, 0.1572]
Continuous-time GRU	0.4826	0.0325	[−0.0118, 0.0938]	0	0.3388	0.0684	[0.0135, 0.1148]
Continuous-time GRU with ρ_t	0.4994	0.0157	[−0.0713, 0.0844]	0.8852	0.3629	0.0443	[−0.0060, 0.0852]
Mamba SSM	0.4859	0.0292	[−0.0229, 0.0794]	0	0.3484	0.0588	[0.0100, 0.1023]
Mamba SSM with ρ_t	**0.4796**	**0.0355**	[−0.0476, 0.1096]	0.8682	0.3401	0.0671	[0.0254, 0.1061]

The correlated Mamba-style selective state-space model has the best average $\mathbb{Q}$ score. The diagonal continuous-time GRU is close: its score of 0.4826 is slightly better than the diagonal Mamba score of 0.4859, but their paired improvement is only 0.0033, with a 95% interval of

$[-0.0483, 0.0663]$. The GRU therefore provides a competitive continuous-time memory parameterization, not a statistically established improvement. Relative to its diagonal version, however, its mean improvement is only 0.0063, with a direct paired 95% interval of $[-0.0429, 0.0484]$. The extra correlation state is therefore learnable but does not yet deliver a statistically established predictive gain. The correlated LSTM is materially worse in $\mathbb{Q}$, especially in the third test block, which is direct evidence of overfitting. The LSTM improvement in the attainable $\mathbb{P}$-drift target remains the clearest result. The GRU correlation head also deteriorates its diagonal model: the direct $\mathbb{Q}$ improvement is -0.0168, with a 95% interval of $[-0.1325, 0.0637]$, and its attainable-$\mathbb{P}$ error increases from 0.3388 to 0.3629. Thus an exponential memory gate alone does not resolve the spot-correlation overfitting problem. These diagnostics evaluate the dynamics model; final model selection must be based on out-of-sample hedging performance for economically relevant portfolios.

2.1.2 Observable spot–ATM-volatility correlation

The factor score evaluates the latent diffusion, whereas a directly observable leverage diagnostic is the expiry-dependent correlation

$$\rho_{S,\mathrm{ATM}}(\tau) := \mathrm{Corr}\left(\frac{\Delta S_t}{S_{t-1}}, \Delta \log \sigma_{\mathrm{ATM},t}(\tau)\right), \tag{15}$$

where spot returns are forward-adjusted and $\sigma_{\mathrm{ATM},t}(\tau)$ is the Black implied volatility at a fixed rolling business-day tenor τ. The historical curves in Figure 1 use the published SANOS ATM term structure. The full curve contains 890 daily transitions from January 2022 to August 2025; the recent curve contains the 258 transitions from 13 August 2024 onward.

For the model curves, we start from the calibrated state on 29 August 2025 and simulate one business day under $\mathbb{Q}$ using eight DLV substeps and 5,000 paths arranged as 2,500 antithetic pairs. The constant model keeps the unconditional factor loadings, while the Mamba-style selective SSM applies its state-dependent factor amplitudes. At the next date, ATM prices are inverted to Black volatilities at the same rolling tenor. When that tenor lies between two surviving fixed expiries, total variance is interpolated linearly. The 125D simulation point is omitted because a one-day rolling 125D volatility requires the birth of an expiry beyond the terminal model date.

Table 2: Selected points from Figure 1. The model columns are total observable correlations, including the DLV and 5% Black background contributions.

Business days	Historical	Recent	Constant $\mathbb{Q}$	Mamba SSM $\mathbb{Q}$
1	−0.382	−0.510	−0.619	−0.667
5	−0.689	−0.790	−0.589	−0.653
10	−0.766	−0.861	−0.645	−0.714
20	−0.806	−0.894	−0.652	−0.731
30	−0.762	−0.881	−0.601	−0.678
60	−0.752	−0.869	−0.597	−0.651
90	−0.742	−0.856	−0.584	−0.623
120	−0.733	−0.845	−0.567	−0.596

The selective backbone improves the root-mean-square discrepancy from the full historical curve from 0.148 to 0.098. It is close to the full-sample 5D correlation (-0.653 versus -0.689), but does not reproduce the stronger recent value of -0.790. More generally, it is too negative at

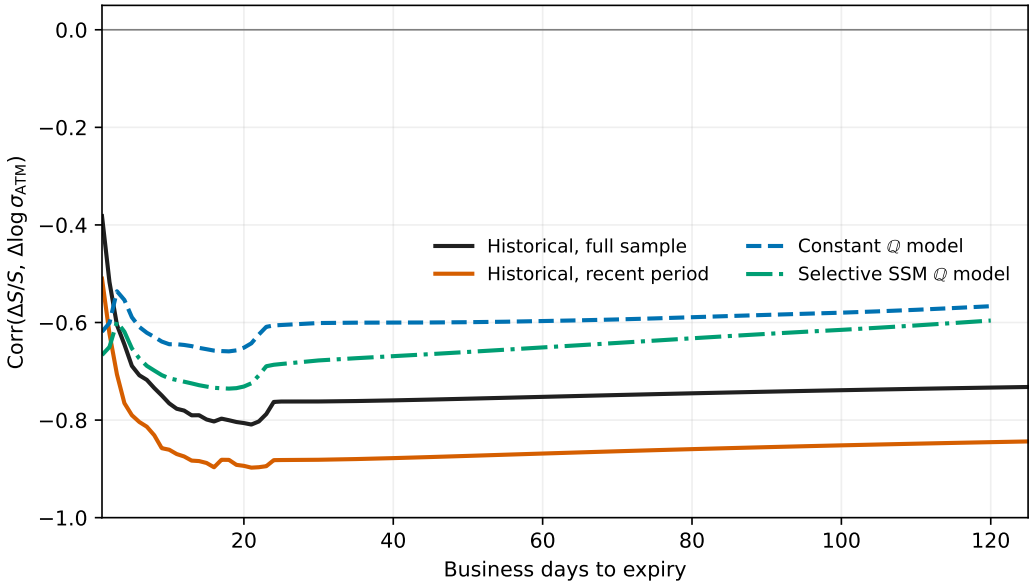


Figure 1: Term structure of daily forward-adjusted spot-return correlation with changes in rolling-tenor ATM Black implied volatility. Historical curves are measured under $\mathbb{P}$; model curves are one-day conditional simulations under $\mathbb{Q}$ from the final calibration state.

1–3D and too weak beyond roughly one month. The corresponding discrepancy from the recent curve is 0.188, compared with 0.246 for the constant model. Thus the selective state improves the observable leverage effect, but the maturity shape and recent regime remain material targets for the state-dependent correlation head in Equation (2).

The 5D gap has a more specific mechanical explanation. During estimation, the first factor is the Gaussian score of the observed product-spot distribution, including the 5% Black background. In the current simulator, that factor settles the residual front density, after which the Black background is sampled independently. This preserves the two marginal laws but not the fitted product-spot copula. On 29 August 2025 the one-day ATM volatility is 9.56%, so the independent 5% background contributes 27.4% of one-day spot variance. In the corresponding first-order decomposition it attenuates spot correlation by

$$\sqrt{1 - \frac{0.05^2}{0.0956^2}} = 0.852. \tag{16}$$

Applying this attenuation to the recent 5D historical correlation gives $-0.790 \times 0.852 = -0.673$, close to the simulated value of -0.653. The principal 5D discrepancy is therefore the independent product-lift coupling, not the choice between Mamba and GRU memory. A corrected simulator should either bridge the observed product spot directly and sample the latent components conditionally, or estimate and simulate a genuinely latent spot score. The learned ρ_t head must then be propagated into the path covariance; it is not used by the present constant/selective curves. The remaining short- and long-maturity shape differences are separate loading-basis and state-dependence questions.

2.1.3 Background-factor identification and estimates

The product representation does not identify a unique latent background process. At one expiry, many joint laws of (X_T, Y_T) produce the same law of $S_T = X_T Y_T$. Option prices therefore identify the product law but not the allocation of its risk between the discrete factor and the smoothing factor. We distinguish three specifications rather than treating them as mutually identified estimates.

First, the production baseline fixes

$$\frac{\mathrm{d}Y_t}{Y_t} = \sigma_Y \,\mathrm{d}B_t, \qquad \sigma_Y = 5\%, \tag{17}$$

with B independent of the DLV factors. This is a modeling convention, not an estimated process. It retains the simple Black convolution and the current closed risk-neutral drift equations.

Second, the released SANOS state provides an observable model-component proxy for a floating background: its Black-kernel total-variance curve. We retain the published smoothness multiplier and interpolate total variance to the GROMM2 expiry grid. Figure 2 summarizes its history. Median annualized kernel volatility is between 7.35% and 7.57% across the grid, compared with the fixed 5% convention, and its 10–90% range at 5D is 4.86%–13.20%. The first kernel-volatility factor explains 77.6% of daily log-change variance; two factors explain 90.3%, four explain 96.3%, and eight explain 98.6%.

Third, as a parsimonious stochastic-volatility benchmark, we fit under $\mathbb{P}$ the one-factor CIR/Heston variance law [Hes93]

$$\mathrm{d}v_t = \kappa_P(\theta_P - v_t)\,\mathrm{d}t + \nu\sqrt{v_t}\,\mathrm{d}B^v_t \tag{18}$$

to the released one-business-day kernel variance, using the exact CIR conditional first two moments in a Gaussian quasi-likelihood. The estimates are $\kappa_P = 19.74$, $\sqrt{\theta_P} = 9.36\%$, and $\nu = 0.935$. The Feller ratio $2\kappa_P\theta_P/\nu^2$ is 0.395, so a positivity-preserving scheme would be required. The standardized variance innovation has correlation -0.320 with the observed product-spot score; a ten-day moving-block bootstrap gives the 95% interval $[-0.413, -0.207]$. This last number is a reduced-form observable correlation, not an identified correlation between the unobserved Y return and its variance shock.

We also test an independent Heston background directly in the surface calibration. On 29 August 2025, a Heston model is fitted to the SANOS surface, scaled to 5% instantaneous volatility, and its correlation is set to $\rho = -0.612$. The resulting background has material downside skew. At representative strikes $K_z = \exp\{z\, 9.56\%\sqrt{T}\}$, its left-minus-right implied-volatility spread between $z = -1$ and $z = 1$ is 2.12, 2.59, 2.61, and 2.51 volatility points at 5D, 20D, 60D, and 120D, respectively. This is a reallocation of the observed skew between the background and residual DLV components: recalibration still targets the same total option surface.

For a historical comparison, we keep the fitted Heston parameters fixed, allow its initial variance to follow the scaled released one-day kernel variance, and recalibrate the rank-200 residual DLV history using the corresponding date-specific Heston pricing kernels. Table 3 compares this rolling-kernel extraction with the otherwise identical 5% Black extraction. Heston does not improve the fit: the fraction inside spread falls by 0.48 percentage points, while mean excess outside spread and midpoint error both increase. This is a like-for-like comparison on the new six-month synthetic book: 7,547,934 SANOS-generated quotes over 45 expiries. It is not directly comparable with the earlier affine-200 result of 95.02% inside bid–ask, which used 2,192,458 retained market quotes on the original 25-daily-expiry grid.

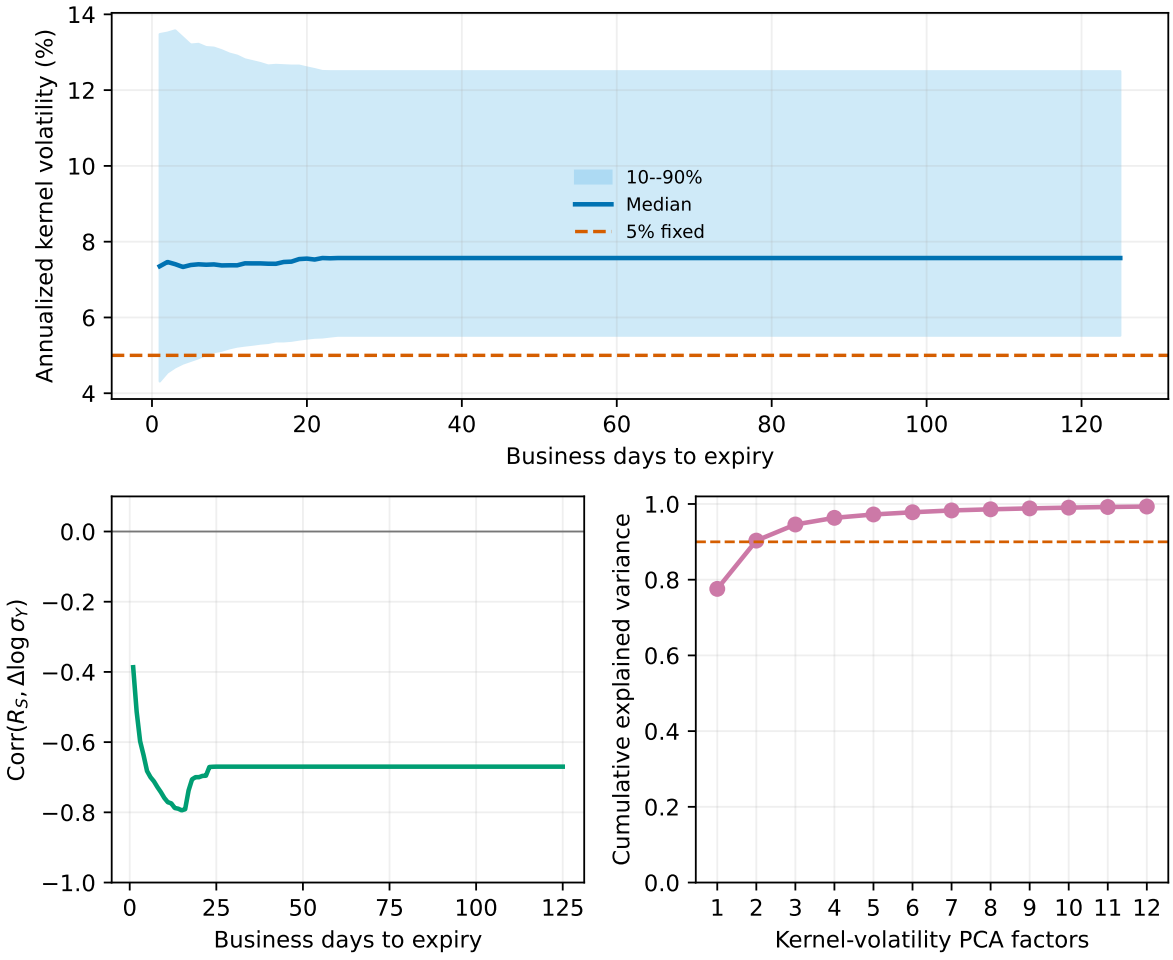


Figure 2: Historical estimates for the released SANOS Black-kernel process. Left: the kernel-volatility term structure. Centre: its daily correlation with forward-adjusted spot returns. Right: cumulative PCA power of daily log-kernel-volatility changes. Beyond the released terminal kernel node, the curve follows the documented flat annualized-volatility extrapolation.

Table 3: Full-history residual-DLV calibration under Black and Heston background kernels on the same six-month synthetic SANOS book. Errors are measured in synthetic half-spreads.

Background	Inside spread	Mean excess	Midpoint MAE
Black, fixed 5%	83.59%	0.170	0.635
Heston, date-specific variance	83.11%	0.270	0.755

More importantly, the rolling extraction does not separate Heston variance from residual DLV risk. After removing the estimated front factor, the Heston variance score still explains 23.0% of residual DLV innovation energy. Its largest absolute correlations with the retained DLV factors are 0.767 in training and 0.668 on the chronological holdout. Thus the Heston kernel adds static skew, but this particular decomposition supplies no evidence of a cleaner dynamic spot–volatility representation.

Table 4: Status of the background specifications. "Estimated" refers to historical $\mathbb{P}$ estimation; none of the stochastic alternatives yet has a separately identified $\mathbb{Q}$ drift.

Specification	Historical result	Risk-neutral status
Independent Black	Fixed $\sigma_Y = 5\%$; not estimated	Current closed construction
Floating SANOS kernel	Median 7.35–7.57%; two PCA factors explain 90.3%	Variance-risk premium not identified
CIR/Heston proxy	$(\kappa_P, \sqrt{\theta_P}, \nu) = (19.74, 9.36\%, 0.935)$	$\mathbb{Q}$ variance drift and latent leverage not identified
Independent Heston pilot	Adds 2.1–2.6 volatility points of downside background skew; 83.11% inside spread historically	Residual DLV innovations remain materially correlated with variance

The floating-kernel and Heston estimates are therefore informative $\mathbb{P}$-models of a background state, but neither experiment identifies the background market price of variance risk separately from the DLV surface. The risk-neutral constructions considered below retain the fixed independent Black background.

2.2 Conditional Arbitrage Check

Let H be the remaining terminal liability of the current option portfolio, h the vector of current hedge holdings, a the post-trade holdings, R the future gains of the hedge instruments, and $C(a-h)$ the cost of changing the hedge. Conditional on the market state $X_t = x$, define the entropic risk of a residual liability Y by

$$\rho_{\gamma,t}(Y) := \frac{1}{\gamma} \log \mathbb{E}[\exp(\gamma Y) \mid X_t = x], \qquad \gamma > 0. \tag{19}$$

The optimal one-period hedge is

$$a^*_{\gamma,t}(H, h, x) := \arg\min_a \rho_{\gamma,t}\Big(H - a^\top R + C(a-h)\Big). \tag{20}$$

Equation (20) defines one hedge rule for arbitrary portfolios, but not one portfolio-independent hedge position. In general, the rule must be conditioned on the remaining liability H. Current cash wealth does not affect the optimizer because exponential utility has constant absolute risk aversion. Current risky holdings h do matter in the presence of transaction costs, position limits, or other trading frictions.

Without trading costs, an interior optimizer satisfies

$$\mathbb{E}\Big[R \exp\Big(\gamma(H - a^{*\top} R)\Big) \mid X_t = x\Big] = 0. \tag{21}$$

The exponentially tilted measure in Equation (21) depends on H. Consequently, hedges of two portfolios do not generally add, and changing the portfolio notional changes the effective risk

aversion. In the frictionless homogeneous case,

$$a^*_{\gamma,t}(cH) = c\, a^*_{c\gamma,t}(H), \qquad c > 0. \tag{22}$$

Thus claim size and risk aversion must be specified together in a hedging benchmark.

There is a useful special case. Suppose that (H, R) is conditionally jointly Gaussian, write $\mu_R := \mathbb{E}[R \mid X_t = x]$, and let $\Sigma_R := \mathrm{Var}(R \mid X_t = x)$ be positive definite. Then

$$a^*_{\gamma,t}(H) = \Sigma_R^{-1} \mathrm{Cov}(R, H \mid X_t = x) + \frac{1}{\gamma}\Sigma_R^{-1}\mu_R. \tag{23}$$

The second term is the position that the optimizer would hold without the liability. Hence the incremental hedge is

$$a^*_{\gamma,t}(H) - a^*_{\gamma,t}(0) = \Sigma_R^{-1} \mathrm{Cov}(R, H \mid X_t = x). \tag{24}$$

It is linear in H and independent of γ. Under $\mathbb{Q}$, where the discounted gains have zero conditional mean, the zero-liability speculative position vanishes. Outside the Gaussian or complete-market setting, Equations (23) and (24) no longer hold in general.

For GROMM, a universal hedging implementation should therefore learn or solve the portfolio-conditioned policy

$$\pi_\gamma(X_t, h_t, \Phi(H))\,, \tag{25}$$

where $\Phi(H)$ is a representation of the remaining portfolio. A natural choice is its vector of discounted payoffs on common GROMM scenarios, possibly compressed into a payoff basis. The reported hedge is the incremental action

$$\pi_\gamma(X_t, h_t, \Phi(H)) - \pi_\gamma(X_t, h_t, 0)\,. \tag{26}$$

This subtraction removes the speculative position under $\mathbb{P}$ and should be zero under $\mathbb{Q}$ in the absence of costs, constraints, and numerical error.

The ultimate performance statistic is the out-of-sample entropic risk of the realized residual hedge loss. For held-out episodes $p = 1, \ldots, P$, with residual losses Y_p, its empirical value is

$$\widehat{\rho}_\gamma(Y) := \frac{1}{\gamma} \log\left(\frac{1}{P}\sum_{p=1}^{P} \exp(\gamma Y_p)\right). \tag{27}$$

Comparisons should use identical market episodes, claim notionals, trading costs, position limits, and risk-aversion levels. Factor-law likelihood is a useful dynamics diagnostic, but improvement in Equation (27) is the economically relevant test.

3 Acknowledgements

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